SUBASH R

AI & Full-Stack Development · Data Science · Cloud (AWS)

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SUMMARY

Third-year Computer Science undergraduate on a dual-degree track: B.E. at REC Chennai (CGPA 9.12, ranked first in the department) and B.Sc. in Data Science at IIT Madras. Three internships across assistive technology, applied AI, and mobile development, and three hackathon wins in 2026. Currently building assistive technology with NIEPMD and AI-assisted study-material pipelines at REC, presented to Tamil Nadu's Minister for AI & IT; earlier shipped a React Native project-management app at IITM Research Park and Power BI analytics over 300,000 records.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering Sep 2024 – May 2028 (Expected)
Rajalakshmi Engineering College (REC), Chennai · CGPA 9.12

Bachelor of Science, Data Science and Applications Sep 2024 – May 2028 (Expected)
Indian Institute of Technology Madras (IIT Madras) · Diploma level completed

TECHNICAL SKILLS

Languages	Python, Java, C, R, SQL, JavaScript, HTML, CSS
Frameworks & Libraries	React, React Native, Next.js, Vue, Node.js, Express, Flask, Jinja2, SQLAlchemy, Bootstrap, JavaFX, JUnit, JWT authentication
Data Science & ML	NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, Seaborn, YOLOv8, U-Net, autoencoders, feature engineering, regression modelling
Databases	PostgreSQL, MongoDB
Cloud & Tools	AWS, Microsoft Azure, Docker, Git, GitHub, Power BI (DAX), Microsoft Excel, Manim, Veo

EXPERIENCE

National Institute for Empowerment of Persons with Multiple Disabilities (NIEPMD), Chennai Aug 2026 – Present
Assistive Technology Intern

- Develop assistive technology for persons with multiple disabilities: taking AudiosenseAI, the SmartAbility Hackathon winning audio-sensing solution, from prototype to a Docker-containerised working build with the institute; details under NDA.

Rajalakshmi Engineering College, Chennai May 2026 – Present
AI Intern, KuttyKoncepts

- Build AI-assisted pipelines that turn faculty lecture notes into tiered, illustrated study materials (Word, PDF, and REC-LMS interactive HTML) for engineering and science courses.
- Create a video-generation engine that turns lesson content into explainer videos, combining HTML templating, Manim, Veo (Google) video generation, AI image generation, and fluid animations.

Shakra Innovations Pvt. Ltd., IITM Research Park Dec 2025 – Jan 2026
Mobile Application Development Intern, Software Team

- Ported the company's web-based Project Management System to a cross-platform React Native mobile app on AWS, with real-time dashboards and task tracking.
- Implemented JWT authentication and role-based access across three user roles.

PROJECTS

Sagar-Nethra, Smart India Hackathon 2026 (PS 26057, Ministry of Earth Sciences) Aug 2026 – Sep 2026
github.com/SUBASH-R-007/SAGAR-NETRA

- Detects marine debris in side-scan sonar imagery using YOLOv8, U-Net, and autoencoder models, with physics-based verification of detections and automated report generation; containerised with Docker.
- Reached mAP50 of 0.819 on real annotated sonar data; the physics-verification stage cut false alarms by 3.2×.

Heavy Material Price Prediction, IIT Madras Capstone Aug 2026

- Regression models that forecast prices of heavy industrial materials from historical market data, built for a Kaggle competition.

AI-Powered, IoT-Enabled Unified Campus App, REC Jan 2026 – Apr 2026

github.com/SUBASH-R-007/ai-smart-campus-app

- Brought eight campus services (canteen, library, academics, transport, attendance, notes, campus map, AI assistant) into one app; user surveys put the drop in average lookup time at around 70%.
- Added RFID login; the built-in AI assistant handles roughly 80% of routine student queries without staff involvement.
- Built in Next.js and containerised with Docker.

Hospital Management System, IIT Madras Nov 2025

github.com/SUBASH-R-007/Hospital-Management-System

- Built a Flask and SQLAlchemy hospital management system with fine-grained role-based access control and a Vue and Bootstrap interface for multiple user types.

Traffic Accident Insights & Trends Dashboard, TANSAM Sep 2025 – Oct 2025

- Built during TANSAM's data analytics training: analysed over 300,000 accident records in Python and R (preprocessing in Excel, SQL on PostgreSQL, feature engineering, ML-based cause analysis) and delivered interactive Power BI dashboards.

LocaLink, REC Oct 2025

github.com/SUBASH-R-007/Localink

- Built a local service marketplace in JavaFX, Java, and PostgreSQL on Microsoft Azure, validated with JUnit functional and integration tests across the core user flows.

ACHIEVEMENTS & AWARDS

Smart India Hackathon 2026, Internal Round Winner Aug 2026

Won with Sagar-Nethra, the sonar-based marine debris detector (see Projects).

SmartAbility Hackathon Winner Aug 2026

Solution now being built with NIEPMD as AudiosenseAI (see Experience).

Presentation to the Tamil Nadu Minister for AI & IT Jul 2026

Presented the KuttyKoncepts internship project to Dr. R. Kumar, Hon'ble Minister for Artificial Intelligence, Information Technology and Digital Services, Government of Tamil Nadu.

KuttyKoncepts AI Hackathon Winner May 2026

Awarded the AI internship at REC (see Experience).

Academic Merit Topper May 2026

Ranked first in the CSE department at REC across all subjects.

IEEE-CS ML Contest Winner Apr 2026

First place among 50+ participants.

CodeSapiens GitHub Contest Winner Aug 2025

Won for code quality and repository structure among entrants from several colleges.

LEADERSHIP & EXTRACURRICULARS

INOVX REC Aug 2025 – Present

Chief Technical Officer

- Lead the club's technical team and build MERN-stack applications used by 100+ students.
- Organised three inter-department coding competitions.

TRAINING & CERTIFICATIONS

Summer Internship and Technical Training Programme, Disaster Management Track (Jun – Jul 2026) India Space Lab

Data Analytics using Power BI, trainee programme (Sep – Oct 2025) TANSAM

AWS Academy Graduate, Cloud Architecting Amazon Web Services

Foundations of Data Science IIT Madras

MongoDB Basics MongoDB University